



Profile

Name: Xianyu Zhu **Email:** zhuxy@mail.ustc.edu.cn
Date of Birth: November 2000 **Website:** zhuxy-cs.github.io/
Field: Computer Science **Advisors:** Hong An and Bin Liu
Skills: C/C++, Python, PyTorch, Linux, CUDA, MPI, NCCL, Verilog
Research Interests: AI Systems, Distributed Training



Education

University of Science and Technology of China (C9, Project 985&211) Sep 2023–Jun 2026

- **GPA:** 3.87/4.3 (90/100) **Degree:** Master' s in Computer Science and Technology
- **Research Group:** Advanced Computer Systems Architecture Laboratory, led by **Prof. Hong An**
- **Selected Coursework:** Foundations of Computer Systems; Design and Analysis of Algorithms; Advanced Computer Architecture; Parallel Programming
- **Research Interests:** High-Performance Computing, Machine Learning Systems, Distributed Training, Communication Optimization
- **Honors and Awards:** First-Class Academic Scholarship, Outstanding Graduates

Northwest A&F University (Project 985&211) Aug 2019–Jun 2023

- **GPA:** 3.77/4.0 (90.5/100) **Rank:** 2/132 **Degree:** B.Sc. in Computer Science and Technology
- **Research Group:** Parallel and Visual Processing Laboratory, led by **Prof. Bin Liu**
- **Selected Coursework:** Data Structures; Operating Systems; Computer Organization and Architecture; Computer Networks; Parallel Computing
- **Research Interests:** Artificial Intelligence, Computer Vision, Object Detection
- **Honors and Awards:** **National Scholarship**; President Scholarship; Outstanding Graduates; Advanced Individual in Social Practice; Excellent College Student;



Publications

High-Performance Computing & Machine Learning Systems

- **First Author:** “Accelerating Large-Scale Distributed Training via Chunk-Wise Gradient Sparsification and Pipelined Communication,” Journal of Computer Science and Technology (JCST), 2026.
- **First Author:** “swPredictor: A Data-Driven Performance Model for Distributed Data Parallelism Training on Large-Scale HPC Clusters,” Performance Evaluation, 2025.
- **Second Author:** “SwFormer: Enabling Faster Foundation Models on New Sunway Supercomputer via Holistic Kernel Tiling and Scheduling,” Journal of Computer Science and Technology (JCST), 2025.

Artificial Intelligence & Computer Vision

- **First Author:** “LAD-Net: A Novel Light Weight Model for Early Apple Leaf Pests and Diseases Classification,” IEEE/ACM Transactions on Computational Biology and Bioinformatics (TCBB), 2023.
- **Second Author:** “Apple-YOLO: A Novel Mobile Terminal Detector Based on YOLOv5 for Early Apple Leaf Diseases,” IEEE International Conference on Computers, Software, and Applications (COMPSAC), 2022.

Research Projects

High-Performance Computing & Machine Learning Systems

- Dataflow Programming and Runtime System Framework Adapted to Domestic Heterogeneous System Architectures (Strategic Priority Program of Chinese Academy of Sciences) **Member**
 - **Developed** a hybrid analytical and data-driven performance model to accurately predict operator and communication overheads in large-scale cluster training.
 - **Designed** a chunk-wise compression algorithm and a pipelined sparse communication framework, achieving $3\times$ to $4.74\times$ communication speedup while reducing peak memory usage.
 - **Extended** the chunk-wise strategy to PowerSGD and sketch-based algorithms, systematically analyzing their feasibility boundaries and operational overheads across different compression paradigms.

Artificial Intelligence & Computer Vision

- AI-Based Early Diagnosis of Apple Leaf Pests and Diseases Using an Inspection Robot (Provincial Innovation and Entrepreneurship Training Program, 2021–2022) **Project Lead**
 - **Proposed** asymmetric dilated convolution and developed LAD-Inception, a multi-scale feature extraction module.
 - **Deployed** the model on an NVIDIA Jetson TX2 and integrated it with a wheeled platform to enable autonomous orchard inspection.
 - **Built** a cloud-edge collaboration workflow for mobile inference, remote monitoring, and result synchronization.
- AI-Based Early Monitoring and Warning of Apple Leaf Diseases and Inspection Equipment Development (Provincial Key Research Program, 2021–2022) **Member**
- Apple Leaf Disease Identification and Detection Using Cloud Collaboration and CNNs (Provincial Innovation and Entrepreneurship Training Program, 2020–2021) **Member**

Experience & Academic Service

- New Media Center, managed content and operations for the official college media **2019–2020**
- Baidu AI Talents Camp, completed machine learning training with PaddlePaddle **Summer 2021**
- Tedu AI Training Program, developed a multi-object annotation platform **Summer 2022**
- Teaching Assistant, “Parallel Computing” **Fall 2022**
- Participant, CCF HPC China 2023 (Qingdao, China) **Aug 2023**
- Participant, Hong Kong PhD Fellowship Summer Workshop (CUHK) **Jul 2025**
- Research Assistant, **Systems and Networking Lab, HKU**, advised by **Prof. Heming Cui**; worked on distributed training systems for multimodal foundation models **Jul–Sep 2025**
- Reviewer, **IEEE Transactions on Parallel and Distributed Systems (TPDS)** **Since 2026**

Competition

- 11th Student RDMA Programming Competition (International Level) **Third Prize**
- 2022 Mathematical Contest in Modeling (MCM) **Meritorious Winner**
- 12th Blue Bridge Cup Competition (Provincial Level) **First Prize**
- 2021 Group Programming Ladder Tournament (National Level) **Third Prize**
- 2022 Chinese Collegiate Computing Competition (Northwest Regional Contest) **First Prize**